www.csvtuonline.com

rinted Pages – 6

337312 (37)

BE (3rd Semester)

Examination, Nov-Dec., 2013

Branch : Mechanical

NUMERICAL ANALYSIS AND COMPUTER PROG. (C & C++)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each question is compulsory. Solve any two parts from (b), (c) and (d).

Q.1. (a) Define absolute, relative and percentage error. 2

(b) Using Newton-Raphson method, find a root of the equation $x \sin x + \cos x = 0$ correct up 3 decimal places, root near to $x = \pi$. 7

55

P.T.O.

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(c) Find a real root of equation $x \log_{10} x =$ by Regula-Falsi method correct to decimal places.

(d) Solve the equations using Gauss-Jord method :

$x + y + z = 9$

$2x - 3y + 4z = 13$

$3x + 4y + 5z = 40$

Q.2. (a) Reduce the pattern $y = ae^{bx}$ where a and are constants, into a linear law of the for $y = mx + c$.

(b) Find the missing values :

x :	0	5	10	15	20	25
y :	6	10	–	17	–	31

www.csvtuonline.com

www.csvtuonline.com

LIBRARY
ACC No
BHIL.AI

(c) If P is the pull required to lift a load W by means of a pulley block, find a linear law of the form P = mW + C connecting P and W, using the following data : 7

P :	12	15	21	25
W :	50	70	100	120

where P and W are taken in kg. wt. Compute P when W = 150 kg.

(d) A slider in a machine along a fixed straight rod. Its distance x cm. along the rod is given below for various values of the time t seconds. Find the velocity of the slider and its acceleration when t = 0.3 second. 7

t:	0	0.1	0.2	0.3	0.4	0.5	0.6
x:	30.13	31.62	32.87	33.64	33.95	33.81	33.24

56

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

Q.3. (a) Write the trapezoidal formula for Numerical Integration.

(b) Attempt any two parts :

Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using

(i) Simpson's 1/3 rule.

(ii) Trapezoidal rule.

(iii) Simpson's 3/8 rule.

(c) Solve $\frac{dy}{dx} = 1 - y$ with y(0) = o in the range $\le x \le 0.3$ by taking h = 0.1 by using modifi Euler's method.

(d) Using Runge-Kutta method of order compute y(0.2) and y(0.4) for $10\frac{dy}{dx} = x$ y^2, y(0) = 1 taking h = 0.2.

www.csvtuonline.com

www.csvtuonline.com

Q.4. (a) Classify the following equations : 2

$$\left(1+x^2\right)\frac{\partial^2 u}{\partial x^2}+\left(5+2x^2\right)\frac{\partial^2 u}{\partial x.\partial t}+\left(4+x^2\right)\frac{\partial^2 u}{\partial t^2}=0$$

(b) Solve $\nabla^2 u = -10(x^2 + y^2 + 10)$ over the square mesh with sides $x = y = 0$, $x = y = 3$ with $u = 0$ on the boundary and mesh length 1 unit. Use Gauss-Seidel method. 7

(c) Evaluate the function satisfying $\nabla^2 u = 0$ at the lattice points given the boundary values as follows. Use iterative method to obtain the solution. 7

(d) Solve $\frac{\partial u}{\partial t}=\frac{\partial^2 u}{\partial x^2}$ in $0 < x < 5$, $t \geq 0$ given that $u(x, 0) = 20$, $u(0, t) = 0$, $u(5, t) = 100$, compute u for the time step with $h = 1$ by crank Nicholson method.

57

www.csvtuonline.com

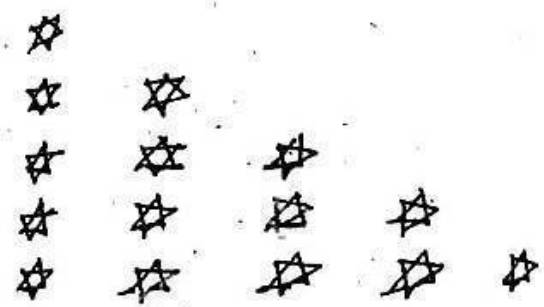
www.csvtuonline.com

www.csvtuonline.com

Q.5. (a) If a is an integer variable $a = \frac{5}{2}$; what value will be return of a. 2

(b) Explain logical operators. 7

(c) Write a program to generate the pattern. 7



(d) Write a program to store 10 numbers in an array and arrange them in ascending order. 7

www.csvtuonline.com

www.csvtuonline.com